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Can AI chatbots help retain customers? An integrative perspective using affordance theory and service-domain logic

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A R T I C L E I N F O

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A B S T R A C T

Chatbots' role in service contexts is changing to ensure better connectedness with customers in the digital marketing era. Thus, developing an understanding of how to enhance user perceptions and behaviors through interface design has become crucial. Using affordance theory and service-dominant logic as theoretical lenses, this study examines how chatbot affordances affect consumers' continuance intention through value-in-use. Data collected from 405 chatbot users of selected banks were analyzed using structural equation modeling. The results reveal that chatbot affordances positively influence personalization and experience. Furthermore, personalization influences switching costs and psychological ownership, whereas experience affects psychological ownership. Moreover, switching costs affect continuance intention through inattentiveness to alternatives. Finally, psychological ownership influences customer citizenship behavior, which impacts continuance intention.

1. Introduction

The arrival of big data and advancements in machine learning have transformed artificial intelligence (AI) into a buzzword, and it not only has become a primary focus of many businesses and organizations, but also has changed their operations (Alabed et al., 2022; Baabdullah et al., 2022). One clear illustration of AI technologies is conversational agents. Depending on the communication mode, conversational agents can be classified as text- or voice-based. Text-based conversational agents often are referred to as chatbots, and they interact with users via text messages, whereas voice-based conversational agents are called digital voice assistants, and they allow users to interact through voice input (Aw et al., 2022). Chatbots' role in service contexts is changing to ensure better connectedness with customers in the digital marketing era (Kautish et al., 2023).

Previous studies have assessed chatbot use from different perspectives. Some examined adoption or behavior intention when using chatbots. These studies confirmed the key roles of trust (Chen et al., 2023; Song et al., 2022a, 2022b), satisfaction (Hari et al., 2022; Poushneh, 2021; Prentice et al., 2020), and enjoyment or playfulness (Yen and Chiang, 2020) in chatbot use. In this regard, the technology acceptance model (Belanche et al., 2019; Kim et al., 2020), unified theory of acceptance and use of technology (Moriuchi, 2021), and other studies derived from these models have gained popularity as a theoretical foundation for examining users' acceptance of chatbots. Others have applied the computers-are-social-actors (CASA) paradigm (Reeves and Nass, 1996) to investigate whether humans interacting with computers exhibit social behaviors similar to those observed in human-to-human communication (Belanche et al., 2020; Tan and Liew, 2020). They have argued that conversational cues with human-like attributes—e.g., perceived humanness (Belanche et al., 2020), anthropomorphism (Aw et al., 2022), and social presence (De Cicco et al., 2020; Tan and Liew, 2020)—are designed to influence perceptions of chatbots.

Affordances are properties of artifacts that represent action possibilities (Moreno and D'Angelo, 2019). Affordance perception is a cognitive process whereby users perceive possibilities for action while interacting with objects. If chatbots offer ways for customers to achieve goals easily, e.g., getting the latest foreign exchange rates, then chatbots' functional features provide opportunities for customers to interact with them. Furthermore, Sundar et al. (2015) argued that interactive media's technological attributes can function as affordances. If chatbots can be viewed as an interactive medium that, e.g., banks offer to service customers, then chatbots' functional features offer action possibilities for customers. In particular, Wang et al. (2018) indicated that affordance theory provides a comprehensive understanding of the relationship between technology and users. Given that few studies have examined

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how chatbot features determine users' behaviors, the present study adopts affordance theory to offer a comprehensive lens through which to explain how chatbots' functional features facilitate behavioral responses. This leads us to our first research question:

Research Question 1: Do chatbot affordances enhance consumers' continuance intention?

Service-dominant logic contends that service is an input that firms offer to realize the proposed value through the cocreation process with customers (Vargo and Akaka, 2009). Whether customers use a technology service is dependent on perceived affordances of technological functions and use contexts, as meaningful interactions enable sustainable value cocreation (Hsu et al., 2021). In the context of chatbot services, firms do not create chatbot services' value—customers cocreate it. Because consumers interact with AI-enabled applications for problem solving, service-domain logic yields crucial insights into users' perceptions and behaviors concerning their use of chatbot services. *Value-in-use* is defined as value determined by customers while using products or services (Vargo and Akaka, 2009). A holistic assessment of value-in-use would help examine the role of the customer-use process in value creation (Macdonald et al., 2011). Understanding users' perceptions of functions and values is crucial in developing services for value cocreation (Lei et al., 2019). As few studies have examined chatbot services' value-in-use from the service-dominant logic perspective, the present study examines how chatbot affordances influence consumers' continuing use of chatbots through value-in-use, which leads to our second research question:

Research Question 2: How do chatbot affordances affect consumers' continuance intention through value-in-use?

This study investigates chatbot affordances' impacts on consumers' intention to continue chatbot use, with affordance theory and service-domain logic used as theoretical lenses. This study examines how chatbot affordances influence continuance intention through value-in-use in terms of personalization, experience, and relationship. In particular, constraint- and dedication-based relationships were considered based on the dedication-constraint model. This study uses the banking industry as its research context because AI has affected it significantly, and it inherently involves frequent contact with numerous customers (Davenport et al., 2020). The banking industry is viewed as highly competitive because banks worldwide offer similar services with homogeneity (Lappeman et al., 2023). The intense competition is driven by the pursuit of consumer satisfaction, higher interest rates, and technologically advanced design (Suhel et al., 2020). However, manufacturing firms focus on keeping track of supply and ensuring that orders are fulfilled. In particular, chatbots adopted in the manufacturing industry help firms access timely information and prevent extra inventory waste. Unlike the manufacturing industry's use of chatbots to maintain regular supplies of materials or merchandise, banks differentiate themselves by offering customers a better banking experience to outperform competitors through chatbot use.

Economic globalization has led to instability and fragility in the banking industry (Satheesh and Nagaraj, 2021), which has become a pioneer in introducing state-of-the-art technology that delivers superior and quality service to customers. By using advanced technology, a bank can penetrate the market on a larger and wider scale. Banking is one of several industries that have developed chatbot services. Chatbots are used in marketing activities, sales, and customer relationships to boost efficiency and provide personalized services to customers. Furthermore, banks traditionally have been human-powered and have required many humans to implement routine and nonroutine tasks, e.g., making transactions, activating credit cards, applying for loans, and checking account balances. Similar to highly competitive and service-oriented industries, e.g., retail and tourism, frontline employees can focus on value-added nonroutine tasks and allow chatbots to handle repetitive tasks or respond to general and simple customer queries. With the aid of chatbots, banks can use algorithms to predict customers' needs and analyze their preferences to generate better sales or profits; thus, the affordances of chatbots that can offer customers seamless experiences become crucial. Such an understanding will help managers in numerous industries apply AI technologies to exercise digital transformation strategies and improve customer service.

2. Theoretical background

2.1. Types of chatbots

Chatbots have been categorized based on various criteria in previous studies, including physical presence, interaction design, response mechanism, conversation type, and functions. As for physical presence, Araujo (2018) argued that conversational agents can be embodied or disembodied. Embodied conversational agents have a physical presence (body or face) and engage in dialogues using language in real-time communications, whereas disembodied conversational agents communicate with users through messaging or text-based interfaces in a dialogical manner. Typically, chatbots employ text-based interaction (Castillo et al., 2021). Regarding interaction design, Følstad et al. (2019) indicated that chatbot interaction designs are structured on the basis of two dimensions, namely locus of control (chatbot-driven vs. user-driven) and duration of interaction (short-term vs. long-term). Følstad et al. (2019) categorized them into these four groups: customer support (user-driven and short-term); content curation (chatbot-driven and short-term); personal assistant (user-driven and long-term); and coach function (chatbot-driven and long-term).

With regard to response mechanism, Gao et al. (2021) described three response mechanisms that chatbots use: the rule-based model; generative model; and hybrid system. The rule-based model is the simplest form, as it uses predefined sets of responses retrieved from a large collection during conversations with humans, whereas the generative model is more advanced, as it creates new responses and sentences through AI and machine learning. The hybrid system combines the rule-based and generative models, offering partly defined and partly free responses (De Keyser & Kunz, 2022). In terms of conversation type, Shevat (2017) suggested two types of dialogues: task-led conversations and topic-led conversations. The former focuses on a narrow goal orientation to complete a specific task, and the latter sheds light on examining a topic of interest in a more in-depth manner.

As for functions, Almurayh (2021) indicated that chatbots offer information and services to customers and, thus, can be classified into two types: information chatbots and transaction chatbots. Information chatbots help users find information on their queries, whereas transaction chatbots allow users to organize tasks or make decisions efficiently. Considering the aforementioned classifications, the chatbots of interest in this study are disembodied conversational agents with hybrid response mechanisms that can communicate through specialized task-led conversations to help customers find information regarding queries and offer user-driven and short-term customer support.

2.2. Affordance theory

Depending on the research context, affordances' features vary. Treem and Leonardi (2013) identified visibility, editability, persistence, and association as social media affordances, while Rice et al. (2017) suggested six reliable organizational media affordances: pervasiveness; editability; self-presentation; searchability; visibility; and awareness. Moreno and D'Angelo (2019) indicated that identity, social, cognitive, emotional, and functional affordances apply to social media. Chan et al. (2019) proposed four social network site affordances—accessibility, information retrieval, editability, and association—to examine how a user evaluates environmental conditions on social network sites in terms of bullying. Lin et al. (2019) used social commerce affordances—including interactivity, stickiness, and word-of-mouth—to examine the formation of swift guanxi between users and sellers.

Li and Chang (2022) identified five factors—anytime/anywhere connectivity, association, visibility, interactivity, and personalization—as affordances of AI-based customer service from in-depth interviews. Given that Li and Chang (2022) examined AI-enabled chatbot services, similar to this study's context, the study followed this classification and put forth four affordance constructs specifically for chatbots: anytime-anyplace connectivity; information association; visibility; and interactivity. However, personalization was excluded from this study for the following two reasons. First, previous studies have argued that personalization can be a firm's strategy or a customer's perception of a firm's customized service. For instance, Tiihonen and Felfernig (2017) described personalization as customized information that matches customers' specific needs and preferences, whereas Lee and Park (2009) defined personalization as the degree to which a customer feels that the content provided is suitable based on their personal information or whether the content was tailor-made to their needs. In particular, a firm's personalization strategy can influence a customer's perceptions about the degree of service personalization. Furthermore, personalization also is viewed as a subconstruct of value-in-use, according to Ranjan and Read (2016). In the present study, understanding how value is cocreated with customers through chatbot use is more crucial than what the chatbot affords to customers.

Second, customers have access to personalized services only when they actively provide personal information. Due to privacy and security concerns, not every customer prefers to reveal personal data while using chatbot services, e.g., a customer who just checks current exchange rates may not choose to log into the system. Given that using two types of personalization complicates the research framework, personalization was adopted as a subconstruct of value-in-use, instead of chatbot affordances, in this study. In the present study, connectivity means creating a physical link between users and technology artifacts, affording users access to services anytime and anywhere (Fang, 2019). Association refers to the act of establishing relations between individuals or between individuals and information content (Wagner et al., 2014). Visibility is denoted as "the means, methods, and opportunities for presentation" (Bregman and Haythornthwaite, 2001, p. 5), while interactivity refers to the subjective perception of quality interaction between a buyer and seller (Ou et al., 2014).

2.3. Service-dominant logic

Service-dominant logic, proposed by Vargo and Lusch (2004), posits that customers are operant resources or collaborative partners who act on other resources and co-create value with firms. More specifically, good-dominant logic focuses on providing products embedded with value to customers, but service-dominant logic sheds light on value co-creation between customers and firms (Yazdanparast et al., 2010). As a firm cannot create value, it is in a context of its own from a service system perspective (Vargo and Akaka, 2009). Customers derive and determine value contextually in the cocreation process.

Value-in-use represents value that customers determine while using services (Medberg and Grönroos, 2020). In particular, Ranjan and Read (2016) argued that value-in-use presents a customer's experiential assessment of a product proposition, which goes beyond functions or features, corresponding to their motivation, competencies, and actions. Vargo et al. (2008) argued that no value is created until an offering is used based on service-dominant logic. Value-in-use represents all consequences that a customer perceives from the service provider (Macdonald et al., 2016). As an affordance features possibilities for action in the use context (Fang, 2019), it can be interpreted appropriately in value-in-use for chatbot use. As customers engage with chatbot services, their needs can be better met, and value can be cocreated by integrating intended goals with offerings. Integrating the lenses of affordance and service-dominant logic will help examine consumers' continuance intention toward chatbots because a chatbot offers value-added services to customers that enable them to obtain information or solve a problem (Leung and Yan Chan, 2020).

According to Ranjan and Read (2016), three dimensions—experience, personalization, and relationship—constitute value-in-use, and several prior studies have followed this classification, including Fang (2019) and Medberg and Grönroos (2020). Following previous research, this study identified experience, personalization, and relationship as the three subdimensions of value-in-use. Experience is an empathetic and affectional interaction between a chatbot and customer that is perceived as enjoyable and memorable (Song et al., 2022a, 2022b). Personalization is defined as the extent to which service content is perceived to be appropriate based on customers' personal characteristics and specific needs (Lee and Park, 2009; Verhagen et al., 2014). Relationship represents the collaboration and reciprocity between the chatbot and customer (Medberg and Grönroos, 2020), and includes dedication- and constraint-based relationships.

2.4. Dedication- and constraint-based relationship mechanisms

Kim and Son (2009) developed a dual model with two mechanisms, dedication- and constraint-based, to explain postadoption phenomena. They argued that a dedication-based relationship is shaped by satisfying experiences, whereas a constraint-based relationship results from specific investments. These include economic, social, or psychological investments that are difficult to transfer elsewhere, and they eventually create a "locked-in" effect. Kim and Son (2009) suggested that the constraint-based relationship is determined by two service-specific investments, learning and personalizing, in the context of information system services.

Unlike some models that have fixed elements, e.g., perceived ease of use and perceived usefulness in the technology acceptance model, variables adopted in the dedication-based model vary across studies. For instance, Shih et al. (2017) used switching costs and dependency to assess a constraint-based relationship, while cognitive trust and affective trust are viewed as dedication-based relationships. To capture consumers' dependence on chatbot services, this study considered both constraint- and dedication-based relationship mechanisms in the formation of customers' continuance intention.

As Bendapudi and Berry (1997) indicated, constraint creates a "locked-in" effect that reduces opportunities for switching to alternatives, in which both switching costs and attentiveness to alternatives are viewed as constraint-based mechanisms in this study. Li et al. (2018) indicated that switching costs function as behavioral constraints that force individuals to quit pre-existing relationships with retailers. Switching costs constrain consumers' options because the termination of a service relationship incurs losses, e.g., costs or efforts to move data to another service provider. Inattentiveness to alternatives is defined as the extent to which an individual lacks interest in an alternative service provider (Kuem et al., 2017). Users who are not interested in seeking alternatives may choose to maintain a relationship with the current service provider (Kuem et al., 2017). In this study, switching costs represents the costs of switching to another bank, whereas inattentiveness to alternatives is defined as the degree to which customers lack interest in an alternative bank.

A dedication-based relationship focuses on customers' passive participation in terms of psychological ownership and customer citizenship behavior. The concept of ownership then is extended to the marketing domain, in which the potential target of possession can be material or immaterial, e.g., a place or brand (Kumar and Nayak, 2019a, 2019b). Psychological ownership in this study represents consumers' psychological understanding that the target is partly theirs. Furthermore, customer citizenship behaviors demonstrate a relationship affiliation (Chan et al., 2017). According to Chen et al. (2020), the dedication-based mechanism captures the true appreciation of the relationship. As citizenship behaviors describe customers' extra-role behaviors, in which they voluntarily engage in service delivery procedures (Nguyen et al., 2014), they are viewed as dedications that focus on an individual's desire to maintain a relationship with a service provider (Chou and Hsu, 2016). Customer citizenship behavior is defined as extra-role behaviors in which a customer voluntarily performs for enhanced service performance.

2.5. The integration of affordance theory and service-dominant logic

In the banking industry, managers introduce chatbots to complete several tasks to serve consumers, e.g., answering their questions or addressing their concerns. Similar to services performed by traditional frontline employees, customers form perceptions regarding human- to-human interaction. In human-to-chatbot interaction contexts, conversational interfaces shape users' perceptions and determine interaction quality (Go and Sundar, 2019). Essentially, the chatbot serves as a medium or service agent to build relationships with customers. The features of interfaces that afford customers to perform certain actions influence evaluations about interaction quality. Affordances describe the design aspects of the artifacts that a particular capability possessed through the media to trigger a certain action (Sundar, 2008). Thus, the affordance theory lens is adopted in this study to explore potential actions that customers may take to obtain information or solve their problems.

Service-dominant logic was integrated with the affordance theory lens for two reasons. First, banks advance their technological capabilities to serve their customers using AI-enabled services. What AI-enabled chatbots do is not only solve customers' problems functionally, but also establish trust relationships with customers. In particular, the attention of customer service has been shifted from what is exchange to the process of exchange, and from tangible to intangible. A chatbot can be a strategic weapon that helps banks deliver experiential value to customers. More specifically, banks possess service-dominant logic, rather than good-dominant logic, to serve customers because they view customers as exchange partners engaging in the value-creation process.

Second, service-dominant logic sheds light on customers' participation to better meet their needs. In this study's context, chatbots offer 24/7 support to optimize customer experiences, e.g., customers obtain information regarding transaction-related questions or manage accounts easily with a few clicks. When customers actively contact chatbots and engage in human-chatbot interactions, they engage in the value creation process, and value can be co-created in use. Chatbot affordances offer possibilities for customers to take certain actions for task completion, and value is co-created from customers' participation in the service delivery process. Accordingly, this study attempts to integrate affordance theory and service-dominant logic.

3. Hypotheses development

Value is created or realized by customers' use of context and process during service provision (Fang, 2019). Interactions with resources offered by a firm generate value for customers, who function as active participants in integrating skill and knowledge for value cocreation (Bruns and Jacob, 2016). Vo-Thanh et al. (2021) indicated that the congruence of consumers' affordance perceptions and value propositions can generate meaningful interactions, while Gadeikienė et al. (2021) posited that telehealth affords a different value-in-use to stakeholders, e.g., patients. In this study, chatbots demonstrated their capability to interact with users to personalize experiences.

H_1. Chatbot affordances exert a positive influence on personalization.

Value cocreation is a function of interaction because value is created or realized through customers' engagement during the service process (Grönroos and Voima, 2013; Sweeney et al., 2018). Smart technologies can be viewed as tools that enable customer and frontline employee interactions that create value (Marinova et al., 2017). Consumer experience is a key aspect of value creation (Edvardsson et al., 2008), i.e., use of AI-enabled chatbots allows customers to obtain value-in-use. Fang (2019) confirmed that branded apps' affordances, including ubiquitous connectivity and associations, advance users' experiential value. Thus, chatbot affordances influence customers' experiences.

H_2. Chatbot affordances exert a positive influence on experience.

A service provider offers personalized service to try and outmaneuver competitors (Kim and Son, 2009). If customers want to receive the same level of service from alternatives, they must spend time and effort learning the new service's routines or procedures. Indeed, personalization is expected to affect transaction costs (Chen and Hitt, 2002), as it creates switching barriers, or a "locked-in" effect, for customers because they expect a new provider to offer the same level of tailored services (Kim and Son, 2009). Thus, personalization is expected to reduce consumers' time and effort spent searching for information, which decreases switching costs.

H_3. Personalization exerts a positive influence on switching costs.

Personalization means that the chatbot satisfies customers' needs by adapting its dialogue or tasks (Bavaresco et al., 2020). Personalized communication offers useful information to customers and supports their decision making (Van den Broeck et al., 2019). Personalization is viewed as a relative advantage (Cingil et al., 2000) because it involves adjusting the service based on the customer's transaction history and an analysis of customer data (Barry and Charpentier, 2020). If chatbots tailor services to satisfy customers' needs, these personalized services will enhance their fit with the chatbots and increase feelings of possession.

H_4. Personalization exerts a positive influence on psychological ownership.

One goal of conversational agents is to improve customer experiences (Danckwerts et al., 2019). By addressing customers' needs through real-time interactions, chatbots fulfill expectations of the customer experience (Yen and Chiang, 2020). By increasing the feeling of being served at the right moment, chatbots enhance the online consumer experience (Van den Broeck et al., 2019). Abandoning chatbot services may inconvenience customers, requiring additional time and effort to search for substitute services, increasing switching costs.

H_5. Experience exerts a positive influence on switching costs.

An experienced individual tends to feel certain, confident, and comfortable with human-machine communication (Mou et al., 2019). A satisfying experience motivates customers toward greater use over time. According to Leung and Yan Chan (2020), firms use chatbots to enhance the customer experience and, thus, increase customer engagement. Zhao et al. (2016) further argued that customer engagement strengthens the relationship between service providers and users, thereby fostering their sense of ownership and loyalty. Taken together, an experienced customer is likely to interact with chatbots, which increases feelings of ownership. Thus, a positive user experience using chatbots increases individuals' sense of psychological ownership.

H_6. Experience exerts a positive influence on psychological ownership.

Switching costs represent the loss of loyalty benefits when ending a current relationship (Lam et al., 2004). Unlike maintaining a relationship with an existing service provider, establishing a relationship with a new one may incur additional costs. When the price of switching is high, customers may view alternatives as less appealing (Kim and Son, 2009), i.e., once an individual decides to remain with their current provider, they are less likely to examine alternatives. Several previous studies have indicated switching costs' influence on inattentiveness to alternatives, e.g., Li et al. (2018) found that switching costs exert negative influences on customers' interest in alternatives. When customers perceive that the cost of switching to an alternative chatbot service is high, customers will be less interested in seeking alternatives.

H_7. Switching costs exert a positive influence on inattentiveness to alternatives.

Inattentiveness to alternatives refers to the lack of interest in alternative services (Kim and Son, 2009). Xiang et al. (2018) suggested that a dependency relationship is determined by satisfaction with and the quality of the alternative relationship. When an individual is not attracted to alternatives, they tend to maintain a dependency relationship. A chatbot is a strategic, competitive tool that firms use to provide customer service. When customers perceive their provider as superior to competitors, they may not be attracted by alternatives or be interested in seeking viable ones. Therefore, inattentiveness to alternatives enhances consumers' continuance intentions.

H_8. Inattentiveness to alternatives exerts a positive influence on continuance intention.

Psychological ownership denotes an individual's possessive tendencies (Morewedge and Giblin, 2015), and it gives rise to positive motivational and behavioral outcomes toward the target object, regardless of whether an individual actually possesses the target (Kumar and Nayak, 2019a, 2019b). In particular, a feeling of possession enhances an individual's sense of responsibility, which elicits positive behaviors (Buchko, 1992). Customer citizenship behaviors are viewed as value-creation behaviors that motivate them to perform extra-role behaviors (Yi & Gong, 2013). Furthermore, Guo et al. (2016) indicated that psychological ownership positively enhances engagement behaviors. Zhang and Xu (2019) suggested that destination psychological ownership positively affects residency citizenship behaviors in terms of positive word-of-mouth, helping behaviors, supporting behaviors, protective behaviors, and tolerant behaviors. When customers have a feeling of possession regarding chatbots, they tend to behave voluntarily and discretionarily to help the chatbots improve performance.

H_9. Psychological ownership exerts a positive influence on customer citizenship behavior.

In the organizational literature, Ma et al. (2016) found that positive emotion, continuance commitment, and workplace social inclusion mediate the influence of employees' organizational citizenship behavior on turnover intention. Mandl and Hogreve (2020) argued that customer citizenship behavior—in terms of feedback, advocacy, helping, and tolerance—increases customers' repurchase behaviors. In the context of chatbot service, customers who display customer citizenship behavior may share their positive experiences with others and write positive reviews about using chatbots. They may even provide useful suggestions on how the service can be improved. When customers are willing to create positive word-of-mouth and promote chatbot services, they tend to continue using chatbot services.

H_{10}. Customer citizenship behavior exerts a positive influence on continuance intention.

4. Methodology

4.1. Research setting and data collection

Banks that use chatbots with customers provide an appropriate context in which to examine the proposed model (Fig. 1). Data for this study were collected through an online questionnaire survey. Before data collection, two experts who have worked at a bank for >10 years and three professors in the marketing strategies field were asked to analyze the questionnaire's clarity and fluency. A pretest was conducted on 30 students majoring in business administration to examine the survey's reliability and validity. Following the pilot test, minor changes were made to the items' wordings. Google Forms, a free Google application, was used to create the questionnaires.

This study targeted customers from Taiwanese banks known for their AI chatbot services, namely E. Sun Commercial Bank, Chinatrust Commercial Bank, Fubon Bank, Taishin International Bank, and Cathay United Bank. The main questionnaire was distributed through a link to the online survey, which also was posted on several popular forums and fan pages about bank services or chatbots on Facebook and Instagram. Respondents who have used chatbot services at least once were asked to answer a series of research questions. Altogether, 415 respondents who have used chatbot services were invited to answer the questionnaire, with 10 incomplete responses deleted; thus, 405 usable responses were retained. The distribution of gender was relatively average, with 66.3 % of the respondents being female. Most of the respondents were under age 30 (33.5 %), followed by ages 31 to 40 (30.6 %). Furthermore, 56.5 % held a bachelor's degree, and 22.2 % held graduate degrees.

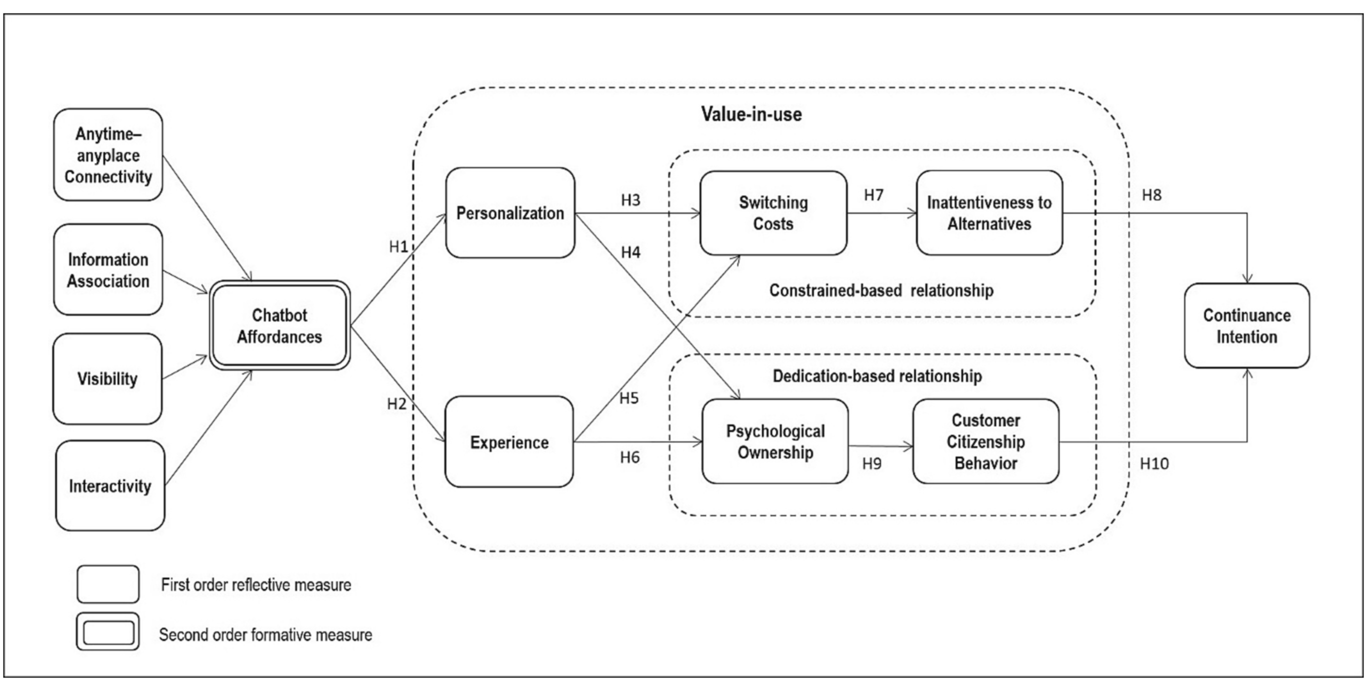

Fig. 1. The research framework of this study.
Note: *$p < 0.05$, **$p < 0.01$ ***$p < 0.001$.

4.2. Measures

This study investigated chatbot affordances' effects on continuance intention through value-in-use (personalization, experience, constraint- and dedication-based relationships). Two types of measurement models using multiple indicators have been discussed in previous studies: the principal factor (reflective) model and composite latent variable (formative) model (Jarvis et al., 2003). The reflective measurement model denotes manifestations of an underlying latent construct, and causality functions from the construct to the measures. Conversely, the measures determine the latent variable in the formative measurement model, and the measures are causes of the construct, rather than its effects (Diamantopoulos and Winklhofer, 2001).

Jarvis et al. (2003) further identified four decision rules to identify a construct as either formative or reflective: (1) direction of causality from construct to measure implied by the conceptual definition; (2) indicators' interchangeability; (3) covariation among indicators; and (4) construct indicators' nomological net. More specifically, with the reflective relationship, the measures share similar themes and are interchangeable, and the addition or subtraction of an indicator exerts little influence on construct validity. Variation in the indicators is caused by variation in the construct, but not the reverse, and high internal consistency among indicators for theoretical homogeneity (Jarvis et al., 2003).

According to Law et al. (1998), a construct can be multidimensional if it comprises several interrelated dimensions and exists in a multidimensional domain under an overall abstraction. Diamantopoulos et al. (2008) argued that two levels of analysis are necessary when conducting a multidimensional construct. The first level is related to the manifest indicators (first-order) dimension, and the second level entails the individual dimensions to the (second-order) latent construct. Given that chatbot affordances represent action possibilities that chatbots offer customers, it was regarded as a second-order latent construct. Four affordance dimensions—anytime-anyplace connectivity, information association, visibility, and interactivity—were recognized as facets within a multidimensional domain encapsulated by an overarching abstraction. Thus, these affordance dimensions were treated as four separate first-order constructs.

Although each affordance dimension represents facets of chatbot affordances, it could be a separate construct, but remain an integral part of chatbot affordances at a more abstract level. Following Jarvis et al.'s (2003) typology, this study classified chatbot affordances as a Type II (reflective first-order, formative second-order) model in which chatbot affordances become a function of four affordance dimensions. Therefore, chatbot affordances were operationalized as a second-order, formative construct with the four first-order, reflective dimensions of anytime-anyplace connectivity, information association, visibility, and interactivity. However, other constructs are first-order reflective constructs because they are defined as unidimensional and are measured by the questionnaire survey. All the measurement items for each first-order reflective-measured construct were adapted from the previous relevant literature, with minor modifications to suit this study's research context. All items were scored on a seven-point Likert scale, ranging from one, "strongly disagree," to seven, "strongly agree." The appendix A lists each item's content.

Back-translation was conducted to confirm that the items were described correctly in both the English and Chinese versions. As this study collected data from only one source, common method variance could have biased the research findings. To preclude the possibility of common method bias, Harman's single factor test and a collinearity assessment were conducted. For Harman's (1967) single factor test, all items were incorporated into the unrotated exploratory factor analysis. The results found that the first factor explained 38.63 % of the variance, below 50 % (Podsakoff et al., 2003). Similarly, Kock (2015) identified common method bias by assessing variance inflation factors (VIFs) engendered from the full collinearity test. The pathological VIFs ranged from 1.15 to 2.46 for all latent constructs, which were below the 3.3 threshold, indicating that common method bias did not contaminate the research model (Kock and Lynn, 2012).

5. Results

This study adopted partial least squares (PLS)—a latent structural equation modeling (SEM) technique, as implemented in SmartPLS 3.02—for data analysis (Ringle et al., 2015). PLS was appropriate for estimating our research model because it offers a way to estimate component scores directly and, thus, can be used to model formative measured constructs. Under covariance-based analysis, parameter identification problems can be avoided (Bollen, 1989). Thus, PLS-SEM was adopted in this study to measure the research model with the formative measurement construct.

To evaluate the measurement model, content validity, convergent validity, and discriminant validity were tested. Content validity was conducted during the questionnaire design stage by checking for consistency between the literature and measurement items. As for convergent validity, as presented in Table 1, each construct's composite reliability (CR) was higher than 0.7, whereas the ratio of construct variance-to-total variance among indicators (AVE) was above the 0.5 threshold. Furthermore, discriminant validity was assessed by examining the values of the square root of the AVE and the heterotrait-monotrait ratio of correlations (HTMT). The findings indicated that the values of the square root of the AVE were all greater than the interconstruct correlations (off-diagonal entries), and the HTMT values were <0.85, as recommended by Henseler et al. (2015), thereby confirming discriminant validity (Table 2).

Fig. 2 presents the results from the structural model. Chatbot affordances were associated positively with personalization ($\beta = 0.77$, $p < 0.001$), experience ($\beta = 0.73$, $p < 0.001$), and relationship ($\beta = 0.60$, $p < 0.001$), thereby supporting H1 and H2. Furthermore, personalization was related positively to switching costs ($\beta = 0.22$, $p < 0.001$) and psychological ownership ($\beta = 0.24$, $p < 0.001$). Experience exerted a positive effect on psychological ownership ($\beta = 0.30$, $p < 0.001$), but did not exert a significant effect on switching costs ($\beta = 0.13$, $p > 0.05$), thereby supporting H3, H4, and H6, but not H5.

Switching costs were related positively to inattentiveness to alternatives ($\beta = 0.78$, $p < 0.001$), whereas psychological ownership was associated positively with customer citizenship behavior ($\beta = 0.49$, $p < 0.001$). Both inattentiveness to alternatives ($\beta = 0.31$, $p < 0.001$) and customer citizenship behavior ($\beta = 0.52$, $p < 0.001$) directly impacted continuance intention, thereby supporting H7–H10. The structural model accounted for 48 % of the variance in continuance intention, 61 % in inattentiveness to alternatives, 24 % in inattentiveness to customer citizenship behavior, 16 % in switching costs, 16 % in psychological ownership, 60 % in personalization, and 53 % in experience.

6. Discussion, implications, and future research suggestions

6.1. Research findings

This study provides empirical insights into the continued use of chatbot services from the perspective of chatbot affordances. Indeed, a study by Fang (2019) stated that branded apps' affordances create distinctive user experiences and personalization, and build a close relationship with customers. The present study's findings indicate that chatbot affordances—anytime-anyplace connectivity, information association, visibility, and interactivity—significantly influence consumers' value-in-use, including personalization and experience. When chatbots allow customers to connect anytime and anyplace, offer associated information, visualize answers, and facilitate two-way communications, customers tend to perceive customer service as personalized and have positive service experiences.

The most interesting finding for scholars and managers alike might

Table 1
Factor loadings and reliability.

Construct	Loading	T statistics	CR[a]	AVE[b]
First order reflective construct				
Anytime-anyplace connectivity			0.87	0.70
AA1	0.87	41.91		
AA2	0.81	32.90		
AA3	0.82	36.66		
Information association			0.92	0.79
IA1	0.89	64.01		
IA2	0.90	72.60		
IA3	0.88	55.90		
Visibility			0.92	0.79
VI1	0.90	87.88		
VI2	0.89	64.00		
VI3	0.86	48.81		
Interactivity			0.92	0.67
IN1	0.81	38.46		
IN2	0.80	33.36		
IN3	0.73	26.01		
IN4	0.86	64.03		
IN5	0.85	58.78		
IN6	0.84	47.28		
Personalization			0.89	0.68
PE1	0.73	22.97		
PE2	0.88	70.77		
PE3	0.89	68.51		
PE4	0.79	36.87		
Experience			0.93	0.81
EX1	0.86	59.96		
EX2	0.80	29.90		
EX3	0.86	46.90		
Switching costs			0.95	0.8
SC1	0.89	59.02		
SC2	0.92	75.88		
SC3	0.93	92.91		
SC4	0.90	63.36		
Inattentiveness to alternatives			0.96	0.86
IA1	0.91	71.21		
IA2	0.91	72.96		
IA3	0.94	113.63		
IA4	0.95	129.16		
Psychological ownership			0.93	0.78
PO1	0.84	44.98		
PO2	0.89	57.50		
PO3	0.91	79.92		
PO4	0.90	83.12		
Customer citizenship behavior			0.89	0.58
CC1	0.75	23.47		
CC2	0.72	22.46		
CC3	0.75	33.14		
CC4	0.80	41.49		
CC5	0.81	39.97		
CC7	0.72	20.80		
Continuance intention			0.93	0.83
CI1	0.87	54.66		
CI2	0.93	78.42		
CI3	0.93	121.65		
Second-order formative construct				
Chatbot affordances				
Anytime-anyplace connectivity	0.21	20.27		
Information association	0.23	21.49		
Visibility	0.27	29.29		
Interactivity	0.48	36.01		

Note.
[a] CR, composite reliability.
[b] AVE, average variance extracted.

be the two different influences of value-in-use—experience and personalization—in constraint- and dedication-based relationships on continuance intention. This study indicates that personalization exercises positive influences on switching costs and psychological ownership. When personalization satisfies customers' needs, consumers' perceptions of switching costs can be enhanced, and they may have a feeling of possession toward the chatbot. The findings conform to a study by Zhou et al. (2012), who argue that personalization triggers a constraint-based mechanism for commitment. Van den Broeck et al. (2019) posited that chatbots' high degree of personalization is an advanced form of a social cue in human-computer interactions.

Furthermore, experiencing value-in-use exerts positive influences on psychological ownership, a finding that coincides with prior literature that hedonic digital services allow users to form a mental attachment to or feelings of ownership (Fritze et al., 2020). Contrary to our expectations, experience does not impact switching costs. This study demonstrates that the experiential value derived from using chatbot services does not result in switching costs. Experiential value is viewed as an enjoyment-related benefit (Chitturi et al., 2008). Given that customers approach chatbots to obtain fast and reliable information, experiential value can be an additional benefit that may not generate switching barriers.

Underpinned by Kim and Son's (2009) dual model, this research reveals that continued use of chatbots can be achieved through dedication- and constraint-based relationship mechanisms. The findings indicate that for a constraint-based relationship, switching costs enhance inattentiveness to alternatives, whereas for a dedication-based relationship, psychological ownership leads to customer citizenship behavior. Specifically, both dedication- and constraint-based relationship mechanisms motivate customers to continue using chatbot services. These findings confirm chatbot affordances' positive effects on the continued use of chatbot services through a dual model.

6.2. Theoretical contributions

The present study differs from prior research in two critical ways. First, each medium has specific interface features that can shape communications (Sundar et al., 2015). Despite many studies having discussed chatbot user interfaces in general, most focused on technological perspectives, e.g., architectures or processing algorithms (Gnewuch et al., 2018). However, factors that influence chatbot adoption are neglected (Abd-alrazaq et al., 2019). Affordances are viewed as inherent technology functions that hinder or help an individual accomplish a specific goal (Ellmer and Reichel, 2020; Miao et al., 2022). This study identified four key affordance constructs in the context of the banking industry—anytime-anyplace connectivity, information association, visibility, and interactivity—in response to Dong et al.'s (2016) statement that "a context-specific and comprehensive instrument for measuring IT affordances is necessary to address the gap" (p. 2).

From marketing disciplines to social media contexts, the dedication-constraint model demonstrates its merit in explaining individualistic behavior when using services (Gong et al., 2022; Kuem et al., 2017). Several previous studies have demonstrated that dedication- and constraint-based relationship mechanisms may strengthen post-adoption behaviors, e.g., continuance intention (Chen et al., 2020; Kim et al., 2014; Kim and Min, 2015; Zhou et al., 2012). As Kim et al. (2014) suggested, dedication- and constraint-based relationship mechanisms can capture users' information and post-system adoption behavior. This study examined consumers' continued use of chatbot services, so the dedication-constraint model was the overarching theoretical framework used to capture individuals' reactions. The research responds to Kim's (2017) suggestion: "However, a number of studies have concentrated mainly on dedication factors, such as user satisfaction and affective commitment, but the dedication perspective alone is notable to fully capture users' post-adoption decision-making processes" (p. 987).

6.3. Managerial implications

With advances in algorithms, chatbots as a tool for customer service have become increasingly common. This study provides four valid insights for practitioners—not only in banking, but also in other

Table 2
Correlations among major constructs.

Variable	(a)	(b)	(c)	(d)	(e)	(g)	(h)	(i)	(j)	(k)	(l)
(a) Anytime-anyplace connectivity	**0.84**	0.55	0.68	0.68	0.75	0.41	0.28	0.30	0.33	0.57	0.54
(b) Information association	0.45	**0.89**	0.59	0.66	0.66	0.47	0.26	0.28	0.37	0.51	0.53
(c) Visibility	0.56	0.52	**0.89**	0.80	0.76	0.70	0.39	0.44	0.58	0.57	0.65
(d) Interactivity	0.58	0.59	0.71	**0.82**	0.81	0.61	0.41	0.46	0.58	0.62	0.62
(e) Personalization	0.61	0.56	0.65	0.71	**0.82**	0.61	0.43	0.47	0.51	0.66	0.64
(f) Experience	0.54	0.56	0.64	0.66	0.72	**0.68**	0.40	0.42	0.55	0.73	0.70
(g) Switching costs	0.24	0.23	0.35	0.37	0.39	0.45	**0.89**	0.83	0.54	0.35	0.48
(h) Inattentiveness to alternatives	0.26	0.26	0.40	0.42	0.42	0.55	0.78	**0.93**	0.65	0.39	0.53
(i) Psychological ownership	0.28	0.33	0.52	0.52	0.46	0.69	0.50	0.61	**0.88**	0.54	0.63
(j) Customer citizenship behavior	0.47	0.44	0.49	0.54	0.56	0.47	0.31	0.35	0.48	**0.76**	0.72
(k) Continuance intention	0.45	0.47	0.57	0.56	0.56	0.66	0.44	0.49	0.58	0.63	**0.91**

Note: diagonal elements are the square root of average variance extracted (AVE) of the reflective scales. Off-diagonal elements are correlations between construct. Above the diagonal element are the HTMT value.

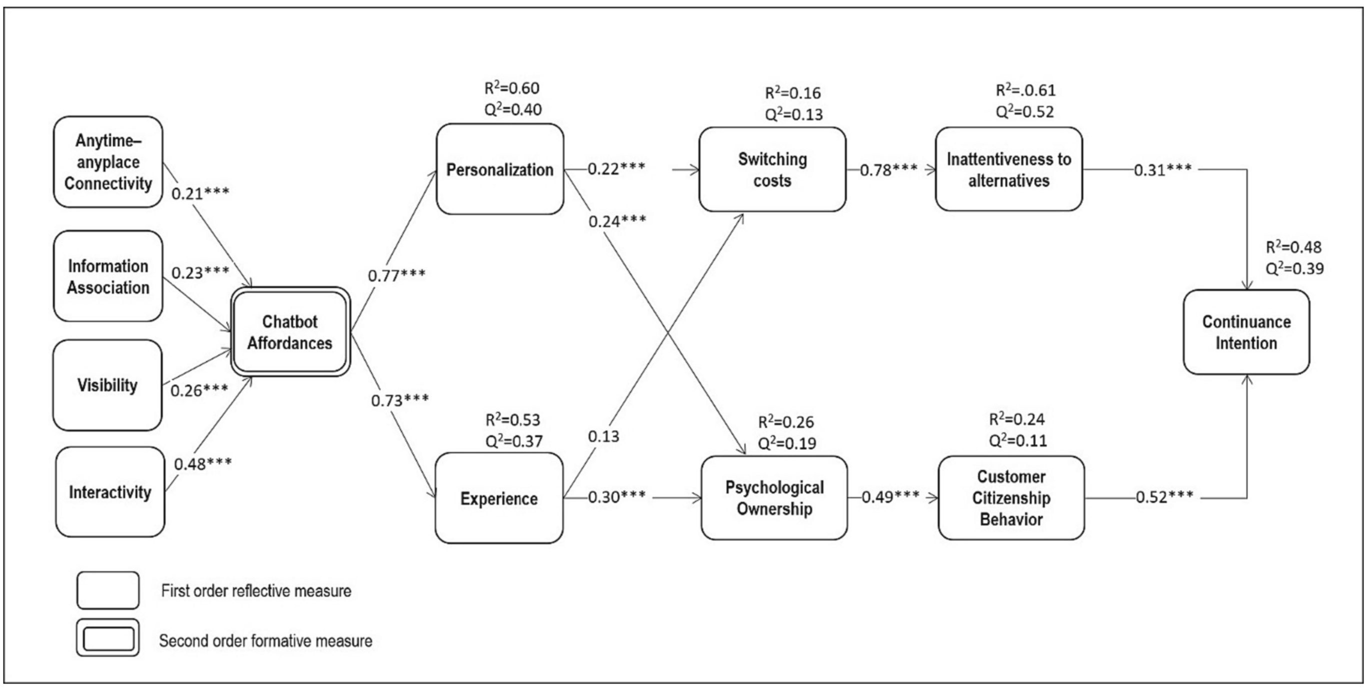

Fig. 2. PLS results for the proposed model.

industries—in the AI era. First, interactivity is the most critical factor for customers' perceived affordances, followed by visibility and information association. Because firms adopt conversational agents to complement human agents, the former function as dialogue partners offering 24/7 support to customers. What customers care about is receiving a timely response and control over the experience, e.g., interaction with frontline staffs. As chatbots depend on natural language processing to understand contexts and respond to customers, gathering more data is needed to improve performance. Presently, chatbots cannot address complex requests or implement nonroutine tasks (De Cicco et al., 2020). Inefficiency or misunderstanding weakens customers' willingness to interact with chatbots, which require customer use for extensive training and to reduce errors. As such, managers may provide incentives for customers to accumulate data so that chatbots can answer complicated questions and let users lead the questions. Furthermore, the visualization of responses is also important for users (Pujiarti et al., 2022). Just as human agents at call centers interact with customers via phone, conversational agents on websites or apps do so via visualized information. Customers experience convenience by copying or recording information for further consideration without pens or paper. Particularly, when time and money to design chatbots for customer services are limited, managers should emphasize interactivity and visibility of chatbot affordances. For instance, managers can introduce gamification mechanisms so that customers can feel like they are exerting control over the functions by clicking buttons, or offer points or coupons to frequent-use customers. Chatbots also respond to customers with visualized cues through augmented/virtual reality technology, so customers can complete tasks easily.

Second, chatbots afford customers personalized services and positive experiences. Customers have access to chatbot services through various interfaces or communication channels—e.g., social media, websites, and messaging apps—so managers can keep track of individual consumer-related preferences by integrating various interfaces and providing services that match consumers' expressed interests. However, personalized services should be used with caution. When customers receive more personalized services, they need to offer relatively more personal data; thus, more personalized customer services elicit more privacy concerns. The balance between thoughtful services and privacy concerns is determined by carefully considering customers' needs and developing firm strategies.

Although the experience of value-in-use does not facilitate switching costs, it functions as a tool for building relationships with customers.

Once chatbot services elicit experiential value, customers may become attached to the chatbot. Feelings of ownership over the chatbot reveal relationship building between customers and firms. Enhancing the customer relationship can be viewed as a powerful chance to upsell (Hildebrand and Bergner, 2019). Given that a chatbot is also a crucial touchpoint of the customer journey, merging online services with offline products is a necessity. Managers must ensure that the customer experience is consistent across various touchpoints.

From the perspective of a constraint-based relationship, personalization that offers utilitarian value to customers enhances switching costs and deters switching to alternatives. As the AI-based chatbot is more cost-effective than humans, consumers' continued use means that firms can allocate resources effectively, in which chatbots perform routine and simple tasks to collaborate with humans. From the perspective of a dedication-based relationship, chatbots are frontline staff that actively interact with customers. This study confirmed that dedication-based relationships (0.52) trigger a stronger influence on consumers' continued use than constraint-based relationships (0.31). Dedication represents an emotion-based evaluation of a long-term relationship, whereas constraint-based relationships focus on the locked-in effect derived from the benefits. In this sense, the chatbot's mission is to maintain long-term relationships with customers. When a dedication-based relationship is difficult to build, the next best thing is to offer functional benefits. Therefore, managers should adopt chatbots to supplement their existing customer interaction platforms for customer relationship management, not merely for cost reduction.

6.4. Limitations and future research directions

This empirical study's findings benefit managers who would like to introduce AI-enabled chatbots for customer service by providing evidence of how they might seek to improve consumers' value-in-use. Nevertheless, the present study is not without limitations. First, the research sample comprised primarily Taiwanese respondents, so the results may not be generalizable to all users in other countries. Thus, this study encourages further research that would include consumers of diverse nationalities to enhance generalizability and overcome this limitation. Furthermore, this study empirically demonstrated affordance theory applicability from the perspective of features in the customer service context, so future research can apply other mechanisms to enhance value-in-use.

Second, switching behavior can be temporary or permanent (Bansal et al., 2005). The form of switching is associated with the properties of the product or service, e.g., as users learn how to use new services and distinguish between old and new services, switching behavior does not necessarily refer to abandoning incumbent services. Therefore, in addition to switching between chatbots at different banks, future research can investigate customers' switching between human agents and conversational agents.

Third, as service chatbots have been adopted in various industries, most undertake and handle similar tasks, e.g., providing information, executing specified transactions, maintaining customer relationships, sending notifications, taking orders, or analyzing data. Although most service chatbots afford similar functions, each chatbot may have specific functions or design interfaces that fit the use context. For example, a robot-advisor can predict future stock market trends, but it is not good at maintaining customer relationships. Therefore, this study's findings may not be generalizable to other research contexts.

Finally, as chatbot affordances in this study are defined as a second-order formative-measured construct, with four first-order reflective constructs—anytime-anyplace connectivity, information association, visibility, and interactivity—the interrelationship among the four first-order reflective constructs is not investigated. As each chatbot affordance dimension represents a facet of chatbot affordances, future studies can examine the interrelationship among chatbot affordance dimensions.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

Data will be made available on request.

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Appendix A. Questionnaire used in this study

Chatbot affordances (adapted from Kim and Son, 2009; Abeele et al., 2017; Rice et al., 2017; Dong et al., 2016; Dong and Wang, 2018)
Anytime-anyplace connectivity
AA1 I have access to the chatbot service and information whenever I need it.
AA2 I have access to the chatbot service and information at all times.
AA3 I have access to the chatbot service and information everywhere I go.
Information association
IA1 The chatbot enables me to find new information I did not know.
IA2 The chatbot enables me to discover new products of which I was unaware.
IA3 The chatbot link me to the information I need.
Visibility
VI1 The chatbot provides me with detailed information containing pictures or videos.
VI2 The chatbot makes products visible to me.
VI3 The chatbot makes visible information about how to solve an encountered problem.
VI4* The chatbot visualizes the answers to my inquiry.
Interactivity
IN1 I had much control over my experience with the chatbot.
IN2 While I talked to the chatbot, I could choose freely what I wanted to see.
IN3 The chatbot facilitated two-way communication between us.
IN4 The chatbot gave me the opportunity to talk back.
IN5 The chatbot responded to my questions quickly.
IN6 I can get information from the chatbot rapidly.

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(*continued*)

Value in use (adapted from Ranjan and Read, 2016)
Personalization
PE1 The benefit, value, or fun of the chatbot service depend on the user and usage condition.
PE2 The chatbot tries to serve the individual needs of each consumer.
PE3 Different consumers, depending on their taste, choice, or knowledge, involve themselves differently in the service.
PE4 The chatbot provides an overall good experience, beyond the "functional" benefit.
Experience
EX1 Using the chatbot service was a memorable experience.
EX2 Depending on the nature of my own participation, my experiences in the process might differ from that of other customers.
EX3 It is possible for a consumer to improve the process by experimenting with and trying new things.
Switching costs (adapted from Trenz et al., 2019)
SC1 Switching to a different chatbot is related to certain hassles.
SC2 It would cost much time and effort to switch chatbot service providers.
SC3 Problems could arise when switching to a different chatbot.
SC4 Switching to a different chatbot is a complex process for me.
Inattentiveness to alternatives (adapted from Li et al., 2018)
IA1 If I need to change chatbot service providers, there are no other good ones from which to choose.
IA2 I would probably be unhappy with the products and services of another chatbot service provider.
IA3 Compared to this chatbot service provider, no others have service with which I would be equally or more satisfied.
IA4 Compared to this chatbot service provider, there are no others with which I would be satisfied.
Psychological ownership (adapted from Fritze et al., 2020)
PO1 It feels as if the chatbot is my service assistant.
PO2 Using the chatbot feels like something that is mine.
PO3 I feel that the chatbot belongs to me.
PO4 I feel a personal connection to the chatbot.
Customer citizenship behavior (adapted from Hur et al., 2018)
CC1 I say positive things about the chatbot to others.
CC2 I give constructive suggestions to the chatbot on how to improve its service.
CC3 The brand of chatbot receives my full support.
CC4 I carefully observe the rules and policies of the chatbot.
CC5 I do things that can make the jobs of the employees easier for the chatbot brand.
CC6 When I have a useful idea on how to improve a service, I communicate it to someone from the chatbot brand.
Continuance intention (adapted from Chen et al., 2020)
CI1 I intend to continue using the chatbot to check information.
CI2 I intend to continue using the chatbot to seek the information I want.
CI3 I intend to continue using the chatbot to seek help.

Note: * represents items that were deleted from the main study because their loadings were small and nonsignificant.

References

Abd-alrazaq, A.A., Alajlani, M., Alalwan, A.A., Bewick, B.M., Gardner, P., Househ, M., 2019. An overview of the features of chatbots in mental health: a scoping review. Int. J. Med. Inform. 132, 103978.

Abeele, M.V., Schouten, A.P., Antheunis, M.L., 2017. Personal, editable, and always accessible: an affordance approach to the relationship between adolescents' mobile messaging behavior and their friendship quality. J. Soc. Pers. Relat. 34 (6), 875–893.

Alabed, A., Javornik, A., Gregory-Smith, D., 2022. AI anthropomorphism and its effect on users' self-congruence and self–AI integration: a theoretical framework and research agenda. Technol. Forecast. Soc. Chang. 182, 121786.

Almurayh, A., 2021. The challenges of using Arabic chatbot in Saudi universities. IAENG Int. J. Comput. Sci. 48 (1), 190–201.

Araujo, T., 2018. Living up to the chatbot hype: the influence of anthropomorphic design cues and communicative agency framing on conversational agent and company perceptions. Comput. Hum. Behav. 85, 183–189.

Aw, E.C.X., Tan, G.W.H., Cham, T.H., Raman, R., Ooi, K.B., 2022. Alexa, what's on my shopping list? Transforming customer experience with digital voice assistants. Technol. Forecast. Soc. Chang. 180, 121711.

Baabdullah, A.M., Alalwan, A.A., Algharabat, R.S., Metri, B., Rana, N.P., 2022. Virtual agents and flow experience: an empirical examination of AI-powered chatbots. Technol. Forecast. Soc. Chang. 181, 121772.

Bansal, H.S., Taylor, S.F., St. James, Y., 2005. "Migrating" to new service providers: toward a unifying framework of consumers' switching behaviors. J. Acad. Mark. Sci. 33 (1), 96–115.

Barry, L., Charpentier, A., 2020. Personalization as a promise: can Big Data change the practice of insurance? Big Data Soc. 7, 2053951720935143.

Bavaresco, R., Silveira, D., Reis, E., Barbosa, J., Righi, R., Costa, C., Silva, E., 2020. Conversational agents in business: a systematic literature review and future research directions. Comput. Sci. Rev. 36, 100239.

Belanche, D., Casaló, L.V., Flavián, C., 2019. Artificial intelligence in FinTech: understanding robo-advisors adoption among customers. Ind. Manag. Data Syst. 119 (7), 1411–1430.

Belanche, D., Casaló, L.V., Flavián, C., 2020. Frontline robots in tourism and hospitality: service enhancement or cost reduction? Electron. Mark. 31 (3), 477–492.

Bendapudi, N., Berry, L.L., 1997. Customers' motivations for maintaining relationships with service providers. J. Retail. 73, 15–37. https://doi.org/10.1016/S0022-4359(97)90013-0.

Bollen, K., 1989. Structural Equations With Latent Variables. John Wiley & Sons, New York.

Bregman, A., Haythornthwaite, C., 2001. Radicals of presentation in persistent conversation. In: Paper Presented at the Proceedings of the 34th Annual Hawaii International Conference on System Sciences, Los Alamitos, CA.

Bruns, K., Jacob, F., 2016. Value-in-use: antecedents, dimensions, and consequences. Mark. ZFP J. Res. Manag. 38 (3), 135–149.

Buchko, A.A., 1992. Effects of employee ownership on employee attitudes: a test of three theoretical perspectives. Work. Occup. 19 (1), 59–78.

Castillo, D., Canhoto, A.I., Said, E., 2021. The dark side of AI-powered service interactions: exploring the process of co-destruction from the customer perspective. Serv. Ind. J. 41 (13-14), 900–925.

Chan, K.W., Gong, T., Zhang, R., Zhou, M., 2017. Do employee citizenship behaviors lead to customer citizenship behaviors? The roles of dual identification and service climate. J. Serv. Res. 20 (3), 259–274.

Chan, T.K.H., Cheung, C.M.K., Wong, R.Y.M., 2019. Cyberbullying on social networking sites: the crime opportunity and affordance perspectives. J. Manag. Inf. Syst. 36 (2), 574–609.

Chen, P.Y., Hitt, L.M., 2002. Measuring switching costs and the determinants of customer retention in internet-enabled businesses: a study of the online brokerage ondustry. Inf. Syst. Res. 13 (3), 255–274.
Chen, J.V., Nguyen, T., Oncheunjit, M., 2020. Understanding continuance intention in traffic-related social media. Internet Res. 30 (2), 539–573.
Chen, Q., Lu, Y., Gong, Y., Xiong, J., 2023. Can AI chatbots help retain customers? Impact of AI service quality on customer loyalty. Internet Res. https://doi.org/10.1108/INTR-09-2021-0686.
Chitturi, R., Raghunathan, R., Mahajan, V., 2008. Delight by design: the role of hedonic versus utilitarian benefits. J. Mark. 72 (3), 48–63.
Chou, S.W., Hsu, C.S., 2016. Understanding online repurchase intention: social exchange theory and shopping habit. IseB 14 (1), 19–45.
Cingil, I., Dogac, A., Azgin, A., 2000. A broader approach to personalization. Commun. ACM 43, 136–141.
Danckwerts, S., Meißner, L., Krampe, C., 2019. Examining user experience of conversational agents in hedonic digital services–antecedents and the role of psychological ownership. J. Serv. Manag. Res. 3 (3), 111–125.
Davenport, T., Guha, A., Grewal, D., Bressgott, T., 2020. How artificial intelligence will change the future of marketing. J. Acad. Mark. Sci. 48 (1), 24–42.
De Cicco, R., e Silva, S.C., Alparone, F.R., 2020. Millennials' attitude toward chatbots: an experimental study in a social relationship perspective. Int. J. Retail Distrib. Manag. 48 (11), 1213–1233.
De Keyser, A., Kunz, W.H., 2022. Living and working with service robots: a TCCM analysis and considerations for future research. J. Serv. Manag. 33 (2), 165–196.
Diamantopoulos, A., Winklhofer, H.M., 2001. Index construction with formative indicators: an alternative to scale development. J. Mark. Res. 38, 269–277.
Diamantopoulos, A., Riefler, P., Roth, K.P., 2008. Advancing formative measurement models. J. Bus. Res. 61 (12), 1203–1218.
Dong, X., Wang, T., 2018. Social tie formation in Chinese online social commerce: the role of IT affordances. Int. J. Inf. Manag. 42, 49–64.
Dong, X., Wang, T., Benbasat, I., 2016. IT affordances in online social commerce: conceptualization validation and scale development. In: Paper Presented at the Twenty-second Americas Conference on Information Systems, San Diego.
Edvardsson, B., Enquist, B., Johnston, R., 2008. Cocreating customer value through hyperreality in the prepurchase service experience. J. Serv. Res. 8 (2), 149–161.
Ellmer, M., Reichel, A., 2020. Mind the channel! An affordance perspective on how digital voice channels encourage or discourage employee voice. Hum. Resour. Manag. J. 31 (1), 259–276.
Fang, Y.H., 2019. An app a day keeps a customer connected: explicating loyalty to brands and branded applications through the lens of affordance and service-dominant logic. Inf. Manag. 56 (3), 377–391.
Følstad, A., Skjuve, M., Brandtzaeg, P.B., 2019. Different chatbots for different purposes: towards a typology of chatbots to understand interaction design. In: St. Petersburg, R., Internet Science: INSCI 2018 International Workshops. Springer International Publishing, pp. 145–156.
Fritze, M.P., Marchand, A., Eisingerich, A.B., Benkenstein, M., 2020. Access-based services as substitutes for material possessions: the role of psychological ownership. J. Serv. Res. 23 (3), 368–385.
Gadeikienė, A., Pundzienė, A., Dovalienė, A., 2021. How does telehealth shape new ways of co-creating value? Int. J. Organ. Anal. 29 (6), 1423–1442.
Gao, C., Lei, W., He, X., Rijke, M., Chua, T.S., 2021. Advances and challenges in conversational recommender systems: a survey. AI Open 2, 100–126.
Gnewuch, U., Morana, S., Adam, M., Maedche, A., 2018. Faster is not always better: understanding the effect of dynamic response delays in human-chatbot interaction. In: Paper Presented at the ECIS 2018 Proceedings at AIS Electronic Library.
Go, E., Sundar, S.S., 2019. Humanizing chatbots: the effects of visual, identity and conversational cues on humanness perceptions. Comput. Hum. Behav. 97, 304–316. https://doi.org/10.1016/j.chb.2019.01.020.
Gong, X., Liu, X., Xiao, Z., 2022. A dedication-constraint model of consumer switching behavior in mobile payment applications. Inf. Manag. 59, 103640.
Grönroos, C., Voima, P., 2013. Critical service logic: making sense of value creation and co-creation. J. Acad. Mark. Sci. 41 (2), 133–150.
Guo, L., Zhang, M., Wang, Y.B., 2016. Effects of customers' psychological characteristics on their engagement behavior in company social networks. Soc. Behav. Personal. Int. J. 44 (10), 1661–1670.
Hari, H., Iyer, R., Sampat, B., 2022. Customer brand engagement through chatbots on bank websites–examining the antecedents and consequences. Int. J. Hum. Comput. Interact. 38 (13), 1212–1227.
Harman, H.H., 1967. Modern Factor Analysis. University of Chicago Press.
Henseler, J., Ringle, C.M., Sarstedt, M., 2015. A new criterion for assessing discriminant validity in variance-based structural equation modeling. J. Acad. Mark. Sci. 43 (1), 115–135.
Hildebrand, C., Bergner, A., 2019. AI-driven sales automation: using chatbots to boost sales. NIM Mark. Intell. Rev. 11 (2), 36–41.
Hsu, P.-F., Nguyen, T.K., Huang, J.-Y., 2021. Value co-creation and co-destruction in self-service technology: a customer's perspective. Electron. Commer. Res. Appl. 46, 101029 https://doi.org/10.1016/j.elerap.2021.101029.
Hur, W.M., Kim, H., Kim, H.K., 2018. Does customer engagement in corporate social responsibility initiatives lead to customer citizenship behaviour? The mediating roles of customer-company identification and affective commitment. Corp. Soc. Responsib. Environ. Manag. 25 (6), 1258–1269.
Jarvis, C., MacKenzie, S., Podsakoff, P.A., 2003. Critical review of construct indicators and measurement model misspecification in marketing and consumer research. J. Consum. Res. 30 (2), 199–218.
Kautish, P., Purohit, S., Filieri, R., Dwivedi, Y.K., 2023. Examining the role of consumer motivations to use voice assistants for fashion shopping: the mediating role of awe experience and eWOM. Technol. Forecast. Soc. Chang. 190, 122407.
Kim, B., 2017. Understanding key antecedents of user loyalty toward mobile messenger applications: an integrative view of emotions and the dedication-constraint model. Int. J. Hum. Comput. Interact. 33 (12), 984–1000.
Kim, B., Min, J., 2015. The distinct roles of dedication-based and constraint-based mechanisms in social networking sites. Internet Res. 25 (1), 30–51.
Kim, S.S., Son, J.Y., 2009. Out of dedication or constraint? A dual model of post-adoption phenomena and its empirical test in the context of online services. MIS Q. 33 (1), 49–70.
Kim, B., Kang, M., Cho, H., 2014. Determinants of post-adoption behaviors of mobile communications applications: a dual-model perspective. Int. J. Hum. Comput. Interact. 30 (7), 547–559.
Kim, J., Merrill, K., Xu, K., Sellnow, D.D., 2020. My teacher is a machine: understanding students' perceptions of AI teaching assistants in online education. Int. J. Hum. Comput. Interact. 36 (20), 1902–1911.
Kock, N., 2015. Common method cias in PLS-SEM: a full collinearity assessment approach. Int. J. e-Collab. 11, 1–10.
Kock, N., Lynn, G.S., 2012. Lateral collinearity and misleading results in variance-based SEM: an illustration and recommendations. J. Assoc. Inf. Syst. 13, 546–580.
Kuem, J., Ray, S., Siponen, M., Kim, S.S., 2017. What leads to prosocial behaviors on social networking services: a tripartite model. J. Manag. Inf. Syst. 34 (1), 40–70.
Kumar, J., Nayak, J.K., 2019a. Exploring destination psychological ownership among tourists: antecedents and outcomes. J. Hosp. Tour. Manag. 39, 30–39.
Kumar, J., Nayak, J.K., 2019b. Understanding the participation of passive members in online brand communities through the lens of psychological ownership theory. Electron. Commer. Res. Appl. 36, 100859.
Lam, S.Y., Shankar, V., Erramilli, M.K., Murthy, B., 2004. Customer value, satisfaction, loyalty, and switching costs: an illustration from a business-to-business service context. J. Acad. Mark. Sci. 32 (3), 293–311.
Lappeman, J., Marlie, S., Johnson, T., Poggenpoel, S., 2023. Trust and digital privacy: Willingness to disclose personal information to banking chatbot services. J. Financ. Serv. Mark. 28 (2), 337–357.
Law, K.S., Wong, C.S., Mobley, W.H., 1998. Toward a taxonomy of multidimensional constructs. Acad. Manag. Rev. 23 (4), 741–755.
Lee, E.J., Park, J.K., 2009. Online service personalization for apparel shopping. J. Retail. Consum. Serv. 16 (2), 83–91.
Lei, S.I., Wang, D., Law, R., 2019. Perceived technology affordance and value of hotel mobile apps: a comparison of hoteliers and customers. J. Hosp. Tour. Manag. 39, 201–211.
Leung, C.H., Yan Chan, W.T., 2020. Retail chatbots: the challenges and opportunities of conversational commerce. J. Digit. Soc. Media Market. 8 (1), 68–84.
Li, C.Y., Chang, Z.T., 2022. Chatbots or me? Consumers' switching between human agents and conversational agents. J. Retail. Consum. Serv. 72, 103264.
Li, Y., Liu, H., Lim, E.T., Goh, J.M., Yang, F., Lee, M.K., 2018. Customer's reaction to cross-channel integration in omnichannel retailing: the mediating roles of retailer uncertainty, identity attractiveness, and switching costs. Decis. Support. Syst. 109, 50–60.
Lin, J., Luo, Z., Cheng, X., Li, L., 2019. Understanding the interplay of social commerce affordances and swift guanxi: an empirical study. Inf. Manag. 56 (2), 213–224.
Ma, E., Qu, H., Wilson, M., 2016. The affective and dispositional consequences of organizational citizenship behavior: a cross-cultural study. J. Hosp. Tour. Res. 40 (4), 399–431.
Macdonald, E.K., Wilson, H., Martinez, V., Toossi, A., 2011. Assessing value-in-use: a conceptual framework and exploratory study. Ind. Mark. Manag. 40 (5), 671–682.
Macdonald, E.K., Kleinaltenkamp, M., Wilson, H.N., 2016. How business customers judge solutions: solution quality and value in use. J. Mark. 80 (3), 96–120.
Mandl, L., Hogreve, J., 2020. Buffering effects of brand community identification in service failures: the role of customer citizenship behaviors. J. Bus. Res. 107, 130–137.
Marinova, D., de Ruyter, K., Huang, M.H., Meuter, M.L., Challagalla, G., 2017. Getting smart: learning from technology-empowered frontline interactions. J. Serv. Res. 20, 29–42.
Medberg, G., Grönroos, C., 2020. Value-in-use and service quality: do customers see a difference? J. Serv. Theory Pract. 30 (4/5), 507–529.
Miao, Y., Du, R., Ou, C.X., 2022. Guanxi circles and light entrepreneurship in social commerce: the roles of mass entrepreneurship climate and technology affordances. Inf. Manag. 59, 103558 https://doi.org/10.1016/j.im.2021.103558.
Moreno, M.A., D'Angelo, J., 2019. Social media intervention design: applying an affordances framework. J. Med. Internet Res. 21, 11014. https://doi.org/10.2196/11014.
Morewedge, C.K., Giblin, C.E., 2015. Explanations of the endowment effect: an integrative review. Trends Cogn. Sci. 19 (6), 339–348.
Moriuchi, E., 2021. An empirical study on anthropomorphism and engagement with disembodied AIs and consumers' re-use behavior. Psychol. Mark. 38 (1), 21–41.
Mou, Y., Xu, K., Xia, K., 2019. Unpacking the black box: examining the (de)gender categorization effect in human-machine communication. Comput. Hum. Behav. 90, 380–387.
Nguyen, H., Groth, M., Walsh, G., Thorsten, H.T., 2014. The impact of service scripts on customer citizenship behavior and the moderating role of employee customer orientation. Psychol. Mark. 31 (12), 1096–1109.
Ou, C.X., Pavlou, P.A., Davison, R.M., 2014. Swift guanxi in online marketplaces: the role of computer-mediated communication technologies. MIS Q. 38, 209–230.

Podsakoff, P.M., MacKenzie, S.B., Lee, J.Y., Podsakoff, N.P., 2003. Common method biases in behavioral research: a critical review of the literature and recommended remedies. J. Appl. Psychol. 88 (5), 879–903.
Poushneh, A., 2021. Humanizing voice assistant: the impact of voice assistant personality on consumers' attitudes and behaviors. J. Retail. Consum. Serv. 58.
Prentice, C., Dominique Lopes, S., Wang, X., 2020. The impact of artificial intelligence and employee service quality on customer satisfaction and loyalty. J. Hosp. Mark. Manag. 29 (7), 739–756.
Pujiarti, R.N., Lee, B., Yi, M.Y., 2022. Enhancing user's self-disclosure through chatbot's co-activity and conversation atmosphere visualization. Int. J. Hum. Comput. Interact. 1–18.
Ranjan, K.R., Read, S., 2016. Value co-creation: concept and measurement. J. Acad. Mark. Sci. 44 (3), 290–315.
Reeves, B., Nass, C., 1996. The Media Equation: How People Treat Computers, Television, and New Media Like Real People and Places. Cambridge University Press, New York.
Rice, R.E., Evans, S.K., Pearce, K.E., Sivunen, A., Vitak, J., Treem, J.W., 2017. Organizational media affordances: operationalization and associations with media use. J. Commun. 67 (1), 106–130.
Ringle, C.M., Wende, S., Becker, J.M., 2015. SmartPLS 3, from. https://www.smartpls.com/.
Satheesh, M.K., Nagaraj, S., 2021. Applications of artificial intelligence on customer experience and service quality of the banking sector. Int. Manag. Rev. 17 (1), 9–17.
Shevat, A., 2017. Designing Bots: Creating Conversational Experiences. O'Reilly Media, Boston, MA.
Shih, H.P., Lai, K.H., Cheng, T.C.E., 2017. Constraint-based and dedication-based mechanisms for encouraging online self-disclosure: is personalization the only thing that matters? Eur. J. Inf. Syst. 26 (4), 432–450.
Song, M., Du, J., Xing, X., Mou, J., 2022a. Should the chatbot "save itself" or "be helped by others"? The influence of service recovery types on consumer perceptions of recovery satisfaction. Electron. Commer. Res. Appl. 55, 101199 https://doi.org/10.1016/j.elerap.2022.101199.
Song, X., Xu, B., Zhao, Z., 2022b. Can people experience romantic love for artificial intelligence? An empirical study of intelligent assistants. Inf. Manag. 59, 103595 https://doi.org/10.1016/j.im.2022.103595.
Suhel, S.F., Shukla, V.K., Vyas, S., Mishra, V.P., 2020. Conversation to automation in banking through chatbot using artificial machine intelligence language. In: 2020 8th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO) India. IEEE.
Sundar, S.S., 2008. The MAIN model: a heuristic approach to understanding technology effects on credibility. In: Metzger, M.J., Flanagin, A.J., T. J. D. & C. T. (Eds.), MacArthur Foundation Series on Digital Media and Learning. The MIT Press, Cambridge, MA, pp. 73–100.
Sundar, S.S., Jia, H., Waddell, T.F., Huang, Y., 2015. Toward a theory of interactive media effects (TIME): four models for explaining how interface features affect user psychology. In: Sundar, S.S. (Ed.), The Handbook of the Psychology of Communication Technology. John Wiley & Sons, Inc.
Sweeney, J.C., Plewa, C., Zurbruegg, R., 2018. Examining positive and negative value-in-use in a complex service setting. Eur. J. Mark. 52 (6), 1084–1106.
Tan, S.M., Liew, T.W., 2020. Designing embodied virtual agents as product specialists in a multi-product category E-commerce: the roles of source credibility and social presence. Int. J. Hum. Comput. Interact. 1–14.
Tiihonen, J., Felfernig, A., 2017. An introduction to personalization and mass customization. J. Intell. Inf. Syst. 49 (1), 1–7.
Treem, J.W., Leonardi, P.M., 2013. Social media use in organizations: exploring the affordances of visibility, editability, persistence, and association. Ann. Int. Commun. Assoc. 36 (1), 143–189.
Trenz, M., Huntgeburth, J., Veit, D., 2019. How to succeed with cloud services? Bus. Inf. Syst. Eng. 61 (2), 181–194.
Van den Broeck, E., Zarouali, B., Poels, K., 2019. Chatbot advertising effectiveness: when does the message get through? Comput. Hum. Behav. 98, 150–157.
Vargo, S.L., Akaka, M.A., 2009. Service-dominant logic as a foundation for service science: clarifications. Serv. Sci. 1 (1), 32–41.
Vargo, S.L., Lusch, R.F., 2004. The four service marketing myths: remnants of a goods-based, manufacturing model. J. Serv. Res. 6 (4), 324–335.
Vargo, S.L., Maglio, P.P., Akaka, M.A., 2008. On value and value co-creation: a service systems and service logic perspective. Eur. Manag. J. 26 (3), 145–152.
Verhagen, T., Van Nes, J., Feldberg, F., Van Dolen, W., 2014. Virtual customer service agents: using social presence and personalization to shape online service encounters. J. Comput.-Mediat. Commun. 19 (3), 529–545.
Vo-Thanh, T., Zaman, M., Hasan, R., Rather, R.A., Lombardi, R., Secundo, G., 2021. How a mobile app can become a catalyst for sustainable social business: the case of too good to go. Technol. Forecast. Soc. Chang. 171, 120962.
Wagner, D., Vollmar, G., Wagner, H.T., 2014. The impact of information technology on knowledge creation. J. Enterp. Inf. Manag. 27 (1), 31–44.
Wang, H., Wang, J., Tang, Q., 2018. A review of application of affordance theory in information systems. J. Serv. Sci. Manag. 11 (1), 56–70.
Xiang, L., Zheng, X., Zhang, K.Z., Lee, M.K., 2018. Understanding consumers' continuance intention to contribute online reviews. Ind. Manag. Data Syst. 118 (1), 22–40.
Yazdanparast, A., Manuj, I., Swartz, S.M., 2010. Co-creating logistics value: a service-dominant logic perspective. Int. J. Logist. Manag. 21 (3), 375–403.
Yen, C., Chiang, M.C., 2020. Trust me, if you can: a study on the factors that influence consumers' purchase intention triggered by chatbots based on brain image evidence and self-reported assessments. Behav. Inform. Technol. 1–18.
Yi, Y., Gong, T., 2013. Customer value co-creation behavior: scale development and validation. J. Bus. Res. 66 (9), 1279–1284.
Zhang, H., Xu, H., 2019. Impact of destination psychological ownership on residents "place citizenship behavior". J. Destin. Mark. Manag. 14, 100391.
Zhao, Q., Chen, C.D., Wang, J.L., 2016. The effects of psychological ownership and TAM on social media loyalty: an integrated model. Telematics Inform. 33 (4), 959–972.
Zhou, Z., Fang, Y., Vogel, D.R., Jin, X.-L., Zhang, X., 2012. Attracted to or locked in? Predicting continuance intention in social virtual world services. J. Manag. Inf. Syst. 29 (1), 273–306.

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